GCP vs AWS for your first SaaS app

For an indie developer shipping a portfolio tracker SaaS in 2025, GCP offers the gentler on-ramp while AWS provides the bigger safety net. GCP's Cloud Run is the standout service — true scale-to-zero pricing, one-command deploys, and a generous always-free tier that can keep your backend costs near zero at low traffic. AWS counters with the broadest ecosystem, Hakia a larger community, and services like CloudFront that recently became unbeatable on CDN pricing. The honest truth, though, is that most indie developers in 2025 are finding that a mix of third-party tools (Supabase, Vercel, GitHub Actions) often beats going all-in on either hyperscaler. Here's the full breakdown to help you decide.

What you'd actually deploy on each platform

The first question isn't "which cloud is better" — it's "which services would I actually use?" Both platforms have dozens of compute options, and picking the wrong one is the fastest way to waste a weekend.

On GCP, the answer is straightforward: Cloud Run. Google themselves recommend it over App Engine for new projects. You point it at your GitHub repo or Docker container, run `gcloud run deploy`, and get a live HTTPS URL in minutes. Google Cloud Cloud Run scales to zero when nobody's using your app, which means your bill during those quiet early months can be genuinely close to **$0 for compute**. Cloudchipr For the frontend, Firebase Hosting gives you a global CDN with automatic SSL — deploy with `firebase deploy` and you're done. The whole flow from code to production can take under 30 minutes if you've used Docker before.

On AWS, you have more choices — which is both the advantage and the problem. The three realistic options for a new developer are:

- **App Runner** — AWS's closest equivalent to Cloud Run. Connect your GitHub repo, pick your runtime, and it handles the rest. A lightweight API running ~8 hours/day costs roughly **$25/month**. amazon It's the simplest AWS path, but it lacks Cloud Run's scale-to-zero pricing.
- **Lightsail** — A traditional VPS with fixed pricing. The **$10/month plan** (1 GB RAM, 2 vCPUs) is predictable and simple. Pair it with a **$15/month managed PostgreSQL database** and you have a complete stack for $25/month with a 3-month free trial on both.
- **Elastic Beanstalk** — A PaaS that provisions EC2 instances, load balancers, and auto-scaling behind the scenes. No charge for Beanstalk itself, but the underlying resources (EC2 + ALB) push the realistic minimum to **$30–50/month**. The .ebextensions config files have a real learning curve.

AWS Amplify deserves a mention for frontend hosting — it's excellent for React/Next.js apps with git-push deploys (freeCodeCamp) and a free tier covering **1,000 build minutes and 15 GB served per month**. But for the backend, AWS simply doesn't have a service as elegant as Cloud Run for small-scale deployments.

Realistic costs for an early-stage portfolio tracker

Here's what each platform actually costs once you assemble a complete stack — backend API, PostgreSQL database, file storage, and frontend hosting:

Component	**GCP (Budget)**	**GCP (Comfortable)**	**AWS (Budget)**	**AWS (Comfortable)**
Backend API	Cloud Run: **$0–5**	Cloud Run: **$12**	Lightsail: **$10**	App Runner: **$25–40**
PostgreSQL	Cloud SQL db-f1-micro: **$9–12**	Cloud SQL 1 vCPU: **$37+**	Lightsail DB: **$15**	RDS db.t4g.micro: **$12–15**
Frontend hosting	Firebase Hosting: **$0**	Firebase: **$0**	Amplify: **$0–5**	Amplify: **$0–5**
File storage	Cloud Storage: **$0–1**	Cloud Storage: **$0–1**	S3: **$0–1**	S3: **$0–1**
Monthly total	**$10–18**	**$50+**	**$25–31**	**$40–60**

The database is the biggest single expense on both platforms, and it's worth understanding why. Cloud SQL's cheapest instance (db-f1-micro at ~$8/month) runs 24/7 whether anyone uses your app or not. (CloudWebSchool) AWS RDS is similar — the db.t4g.micro at ~$12/month doesn't scale to zero. This is the cost floor that makes both clouds surprisingly expensive for apps with low initial traffic.

A critical GCP caveat: the db-f1-micro and db-g1-small instances are shared-core machines that **Google explicitly says are "for development only" and aren't covered by their SLA**. (Google) If you want actual production reliability on GCP, the smallest dedicated-core Cloud SQL instance runs about **$37/month —** nearly triple the entry price. (Hamy)

Egress costs (bandwidth for data leaving the cloud) are often cited as a hidden trap, but for early-stage apps they're negligible. AWS now offers **100 GB/month free egress** across all services, and GCP charges about $0.12/GB after a smaller free allowance. A portfolio tracker serving a few hundred users would barely notice.

The real hidden costs to watch for are **load balancers** (~$16–22/month on AWS if you use an ALB, ~$18/month on GCP), **public IPv4 addresses** ($3.60/month on AWS since February 2024), and **orphaned resources —** EBS volumes, unused Elastic IPs, or container images piling up in registries. CloudOptimo Multiple developers report that their AWS bills crept from $50 to $150/month just from forgotten resources.

Free tiers are generous but very different

GCP gives new accounts **$300 in credits** Orthoplex Solutions **valid for 90 days —** enough to thoroughly test a full production stack at zero cost. Google google After that, GCP's always-free tier is genuinely useful: **240,000 vCPU-seconds/month on Cloud Run** (enough for a low-traffic API), google **5 GB of Cloud Storage**, FreeTiers one **free e2-micro VM** perpetually, CloudWebSchool and **2 million Cloud Run requests/month**. google Google Cloud The critical gap is that **Cloud SQL has no free tier —** your database starts costing money from day one after your trial credits expire.

AWS recently overhauled its free tier. Accounts created after July 15, 2025 get a new "Free Plan": **$100 in credits upfront, plus up to $100 more** for completing onboarding tasks, valid for 6 months. DEV Community The key safety feature is that if you exhaust your credits, AWS suspends your account rather than billing you — eliminating the surprise-bill problem that has plagued AWS beginners for years. Cloudwithalon Legacy accounts (created before July 2025) still get the traditional 12-month free tier: DEV Community **750 hours/month of t2/t3.micro EC2**, Ventus Servers Cloudwithalon **750 hours of RDS db.t3.micro**, AWS and **5 GB of S3**. DEV Community

Both platforms offer compelling always-free services. AWS Lambda's **1 million free requests/month** and CloudFront's **1 TB free data transfer** never expire. DEV Community GCP's Cloud Run free tier and the free e2-micro VM are perpetual too. Google Cloud But neither platform lets you run a complete production SaaS — with database — for free indefinitely.

For what it's worth, GCP offers significantly more generous **startup credits**: up to **$200,000** (or $350,000 for AI startups) with no VC or accelerator requirement. google AWS offers up to $100,000 but generally requires an accelerator affiliation. Startupbricks Blog

What developers actually say about the experience

The developer sentiment data from 2024–2026 is remarkably consistent: **GCP wins on developer experience, AWS wins on ecosystem breadth and trust**.

GCP's console is widely described as "cleaner" and "more intuitive." The `gcloud` CLI is praised as one of the best-designed command-line tools in the cloud space. Tech Insider Inventive HQ One developer who spent two years on each platform put it memorably: AWS gives you a wheel, a chunky manual, and the keys, then sends you to twenty different shops to buy the remaining components Medium — while Google looked at that and said, "let's make it simpler." Medium The **Stack Overflow 2024 survey** shows AWS used by **48%** of developers Getint versus GCP's **25%**, Stack Overflow but among developers learning a new platform, the two are nearly tied at ~19% each. Stack Overflow

AWS's overwhelming breadth — **200+ services** Jagadish Writes — is simultaneously its greatest strength and its most common complaint. Eesel AI DEV Community New developers report feeling lost in the console, struggling with IAM permissions, and needing a "PhD in Cloud Economics" to predict their bill. GCP has fewer services but they're more opinionated, which means less decision fatigue and a faster path to deployment. Medium

The "Killed by Google" problem is real, though. Google's history of shutting down products — most recently Google Cloud IoT Core in 2023 — continues to erode trust. The Stack Multiple CTOs and senior engineers explicitly warn against betting on GCP for this reason. A major GCP outage in June 2025 that affected **50+ services across 40+ regions** (impacting Spotify, GitLab, Shopify, and even Cloudflare) didn't help the narrative. ByteByteGo AWS, by contrast, has never killed a major service DEV Community and is widely seen as the "safer" long-term bet.

AWS also wins decisively on **community resources**. There are roughly **230,000 job postings** requiring AWS skills versus ~90,000 for GCP. Certopus Blogs More tutorials, more Stack Overflow answers, more blog posts. If you get stuck at 2 AM, you're statistically more likely to find an AWS answer.

Authentication, databases, and the SaaS essentials

For the specific tools a portfolio tracker needs, the comparison gets interesting — and both platforms have notable weak spots.

Authentication is where GCP has a clear edge. Firebase Authentication offers **50,000 free monthly active users** SuperTokens with drop-in SDKs Logto Firebase that developers consistently call "boring in the best way" — meaning it just works. DEV Community AWS Cognito, by contrast, has some of the worst developer reviews of any AWS service. Tesseral One team reported spending four days fighting Cognito before switching to Auth0 and finishing in four hours. StackShare Cognito's December 2024 pricing overhaul also reduced its free tier from 50,000 to **10,000 MAU** for new user pools. Costgoat For an indie dev, **Firebase Auth is the better native choice**, though third-party options like Clerk (best UI/UX, **10,000 MAU free**) or Supabase Auth (bundled with their database, DEV Community **50,000 MAU free**) DEV Community are strong alternatives on either cloud.

Managed PostgreSQL is expensive on both platforms, and the honest recommendation for most indie developers is to consider a third-party provider instead. AWS RDS and GCP Cloud SQL both start around **$8–15/month** for their smallest instances and quickly climb to $30–50+ for anything production-grade. Hamy Benchmarks from benchANT show RDS delivering better raw OLTP performance, Benchant while Cloud SQL offers more flexible instance sizing. SQLFlash But **Neon** (serverless Postgres with scale-to-zero, free tier with 0.5 GB storage, $19/month Launch plan) and **Supabase** (Postgres + auth + storage PeerDB Blog for $25/month Pro) offer dramatically better value for early-stage apps. Neon's $1 billion acquisition by Databricks in 2025 signals strong long-term backing.

Secrets management tilts toward AWS on the free end: AWS Systems Manager Parameter Store is **completely free** for up to 10,000 parameters at the Standard tier. GCP Secret Manager gives you **6 free active secret versions** and charges $0.06/version/month after that. google CyberAlternatives Both are perfectly adequate for a small SaaS. The practical advice is to not overthink this — environment variables in your deployment platform work fine until you have a reason to need more.

CI/CD is where third-party tools dominate. AWS CodePipeline and CodeBuild are powerful but absurdly complex for an indie developer — configuring IAM roles, trust policies, and multi-service pipelines is enterprise-grade work. Serverless First GCP's Cloud Build is simpler and offers a generous **2,500 free build-minutes/month**. google But **GitHub Actions** (2,000 free minutes/month, massive marketplace, YAML config in your repo) is what most indie developers actually use regardless of cloud platform. PeerSpot If you're deploying a Next.js app, Vercel's zero-config deploys with preview URLs per pull request are in a class of their own.

One area where **AWS clearly wins: CDN and email**. CloudFront's always-free tier includes **1 TB of data transfer and 10 million requests per month** DEV Community — enough for most small SaaS apps. New flat-rate plans launched in late 2025 bundle CDN + WAF + DDoS protection + DNS + SSL Inventive HQ starting at $0/month. GCP's Cloud CDN has no free tier. For transactional email, AWS SES costs just **$0.10 per 1,000 emails** with a free tier of 3,000/month — GCP has no native email service at all, forcing you to use a third-party like Resend or SendGrid.

Conclusion

The practical answer for an indie developer shipping their first production SaaS in 2025–2026 is less about AWS vs. GCP and more about which specific services give you the fastest, cheapest path to production.

If you want the simplest possible cloud-native path, GCP's combination of Cloud Run + Firebase Hosting + Firebase Auth is hard to beat. You can get a full-stack app deployed for **under $15/month** (the database being the main cost), and Cloud Run's scale-to-zero means your compute bill stays near zero during the long tail of early user growth. Cloudchipr The developer experience is consistently smoother, the console is less overwhelming, Orthoplex Solutions and the `gcloud` CLI is genuinely pleasant. Inventive HQ

If you want the broadest ecosystem and the most "safe" long-term bet, AWS has more services, more community support, Dotstark Hakia more job-market relevance, and specific wins on CDN (CloudFront) and email (SES). The new Free Plan's no-surprise-bill design eliminates the biggest historical complaint. But expect to spend more time learning IAM, navigating the console, and configuring services that GCP handles more automatically. DEV Community

The increasingly popular third option: skip the hyperscaler lock-in entirely. A stack of **Supabase** (database + auth + storage, $0–25/month) + **Vercel or Railway** (hosting + CI/CD, $0–20/month) + **Cloudflare** (CDN + DNS, free) + **Resend** (email, free for 3,000/month) gives you everything a portfolio tracker needs for **$0–45/month** with dramatically less complexity than either AWS or GCP. This is what a growing number of indie developers are actually choosing Wowtechub — and for a first production app, that simplicity has real value.